

Verified minimal pairs — frozen π_0 , SIMPLER Bridge

Full per-task layout grid $\times$ 3 reps (n=72 per phrase; basket grid 60 $\rightarrow$ n=180); success = final state at the 60-step horizon; p = two-proportion, top vs. bottom. Highlight marks the changed span.



carrot_on_keyboard

destination noun (family) +18 pp· p=0.0030

25% put the carrot on the black keyboard
25% place the carrot on the keyboard keys
15% place the carrot on the keyboard
7% put the carrot on the keys



carrot_on_wheel

destination noun (family) +25 pp· p=0.0001

31% put the carrot on the black wheel
12% put the carrot on the wheel
6% put the carrot on the rim
6% put the carrot on the tire



carrot_on_ramekin

preposition +19 pp· p=0.0111

40% Place the carrot inside the ramekin.
21% Place the carrot in the ramekin.

object noun +38 pp· p<0.0001

47% Pick up the orange carrot and place it in the white bowl.
10% Pick up the orange vegetable and place it in the white bowl.



coke_can_on_keyboard

object noun +42 pp· p<0.0001

46% Put the red soda cylinder on top of the black keyboard.
4% Put the red soda can on top of the black keyboard.



coke_can_on_plate

object noun +19 pp· p=0.0199

61% put the coke can on the plate
42% put the coke can on the dish

object / destination noun (family) +32 pp· p=0.0001

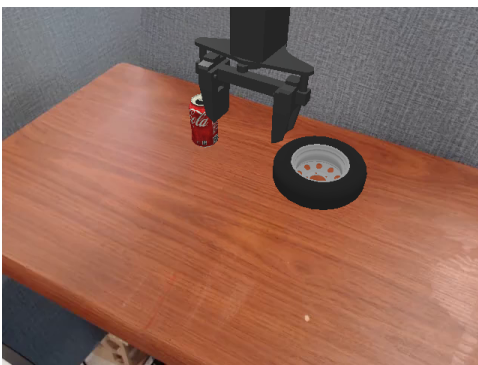
67% put the coke on the plate
61% put the coke can on the plate
53% put the coke can on top of the plate
44% put the coke can on the yellow plate
42% put the coke can on the dish
35% put the can on the plate



coke_can_on_ramekin

destination noun (ladder) +42 pp· p<0.0001

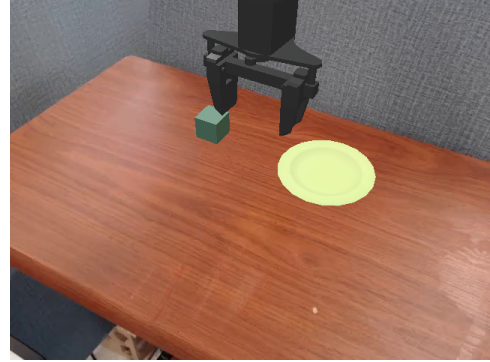
74% place the red cola can inside the white container
69% place the red cola can inside the white basin
69% place the red cola can inside the white bowl
69% place the red cola can inside the white pot
51% place the red cola can inside the white dish
50% place the red cola can inside the white cup
32% place the red cola can inside the white ramekin



coke_can_on_wheel

destination colour +28 pp· p=0.0001

36% Grab the red Coke can and set it on the black wheel.
8% grab the red coke can and set it on the wheel



cube_on_plate

verb +38 pp· p<0.0001

78% place the green cube into the dish
40% put the green cube into the dish

destination colour +30 pp· p=0.0001

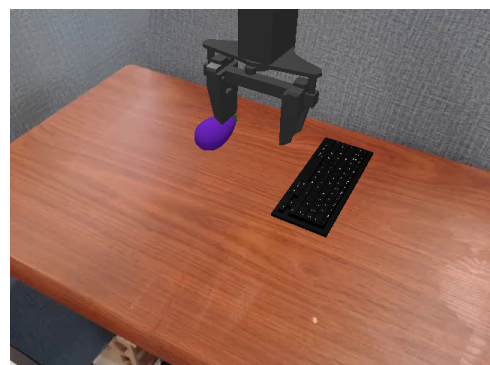
85% Pick up the green cube and put it on the plate.
54% pick up the green cube and put it on the yellow plate

destination colour +18 pp· p=0.0151

82% Place the green block on the plate.
64% Place the green block on the green plate.

object colour +35 pp· p<0.0001

79% put the green block on the plate
44% put the block on the plate



eggplant_on_keyboard

verb +24 pp· p=0.0017

40% Rest the purple eggplant on the black keyboard.

17% Move the purple eggplant on the black keyboard.

articles +19 pp· p=0.0051

32% put eggplant on keyboard

12% put the eggplant on the keyboard

destination colour +42 pp· p<0.0001

54% put the eggplant on the black keyboard

12% put the eggplant on the keyboard

object colour +30 pp· p=0.0002

61% The eggplant needs to end up on the black keyboard.

31% The purple eggplant needs to end up on the black keyboard.

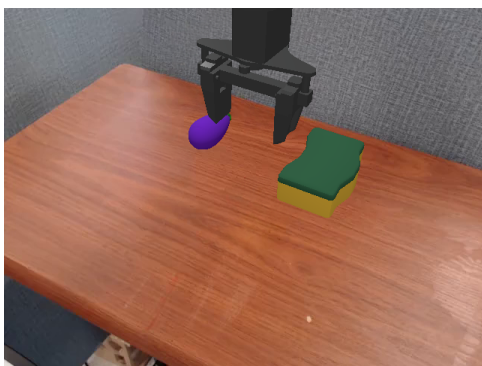


nut_on_plate

object colour +18 pp· p=0.0191

40% drop the nut on the yellow plate

22% drop the grey nut on the yellow plate



eggplant_on_sponge

verb +19 pp· p=0.0051

88% Set the purple eggplant onto the sponge.

68% Move the purple eggplant onto the sponge.

object noun +44 pp· p<0.0001

71% set the purple eggplant on the yellow and green sponge

26% set the purple thing on the yellow and green sponge

word order +49 pp· p<0.0001

85% Set the purple eggplant on the sponge.

36% purple Eggplant set on the sponge.

object colour +61 pp· p<0.0001

69% purple eggplant goes on the sponge

8% eggplant goes on the sponge



nut_on_wheel

verb +17 pp· p=0.0072

25% Put the nut right on the wheel.

8% Position the nut right on the wheel.



pepsi_on_plate

case / punctuation +43 pp· p<0.0001

69% set the pepsi down on the plate

26% Set the Pepsi down on the plate.

case / punctuation +39 pp · p<0.0001

69%	put	the	pepsi	on the	plate
31%	Put	the	Pepsi	on the	plate.

preposition +15 pp· p=0.0419

36% Move the blue can onto the plate.

21% Move the blue can to the plate.

object noun +39 pp· p<0.0001

71% set the pepsi on the plate

32% set the soda on the plate

object colour +38 pp· p<0.0001

64% set the blue coke on the plate

26% set the coke on the plate



spoon_on_towel

colour / preposition (family) +22 pp· p=0.0071

67% put the spoon atop the towel

61% put the spoon on the towel

44% put the spoon on the blue towel



stack_cube

object noun / verb (family) +33 pp· p<0.0001

50% put the green cube on the yellow cube

43% green cube on yellow cube

29% put the green block on the yellow block

25% stack the green cube on the yellow cube

21% stack the green block on the yellow block

17% green block on yellow block